

Displays

D1:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\theta}, \mathbf{w}_k), \quad (1)$$

D2:

$$\mathcal{X}_k = \left\{ \mathbf{x}_k = \mathbf{f}(\cdot, \mathbf{u}_{k-1}, \boldsymbol{\theta}, \mathbf{w}_{k-1}) \circ \cdots \circ \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0, \boldsymbol{\theta}, \mathbf{w}_0) \mid \begin{array}{l} \mathbf{x}_0 \in \mathcal{X}_0, \mathbf{u}_i \in \mathcal{U}_i, \mathbf{w}_i \in \mathbb{W} \\ \boldsymbol{\theta} \in \Theta, \quad i = 1, \dots, k-1 \end{array} \right\}. \quad (2)$$

D3:

$$\tilde{\mathcal{X}}_{k+1} = \{ \mathbf{x}_{k+1} \mid \mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\theta}, \mathbf{w}_k), \mathbf{x}_k \in \tilde{\mathcal{X}}_k, \mathbf{u}_k \in \mathcal{U}_k, \boldsymbol{\theta} \in \Theta, \mathbf{w}_k \in \mathbb{W} \}, \tilde{\mathcal{X}}_0 = \mathcal{X}_0,$$

D4:

$$\mathcal{X} \cap K = \emptyset \implies \mathbb{P}(\mathcal{X}^m \cap K \neq \emptyset \text{ infinitely often}) = 0. \quad (3)$$

D5:

$$\mathcal{X} \cap G \neq \emptyset \implies \mathbb{P}(\mathcal{X}^m \cap G = \emptyset \text{ infinitely often}) = 0. \quad (4)$$

D6:

$$\mathcal{L}^M(\mathbf{z}) = \frac{1}{N} \sum_{k=1}^N \|\mathbf{x}_k(\mathbf{z}) - \mathbf{c}_k^M\|_{\mathbf{Q}_k^M}^2, \text{ with } \mathbf{Q}_k^M = \left(\frac{1}{M-1} \sum_{j=1}^M (\mathbf{x}_k^j - \mathbf{c}_k^M)(\mathbf{x}_k^j - \mathbf{c}_k^M)^T \right)^{-1}, \quad (5)$$

D7:

$$\mathbb{P}(\mathbf{x}^j \in G \text{ i.o.}) \leq \mathbb{P}\left(\bigcup_{m=1}^{\infty} \mathbf{x}^m \in G\right) \implies \mathbb{P}\left(\bigcup_{m=1}^{\infty} \mathbf{x}^m \in G\right) = 1. \quad (6)$$

D8:

$$\mathbb{P}(\mathcal{X}^m \cap G = \emptyset \text{ i.o.}) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} \mathcal{X}^m \cap G = \emptyset\right) = 1 - \mathbb{P}\left(\bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} \mathcal{X}^m \cap G \neq \emptyset\right).$$

D9:

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} \mathcal{X}^m \cap G \neq \emptyset\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} \mathcal{X}^n \cap G \neq \emptyset\right).$$

D10:

$$\mathbb{P}(\mathcal{X}^m \cap G = \emptyset \text{ i.o.}) = 0 \iff \mathbb{P}\left(\bigcup_{n=1}^{\infty} \mathcal{X}^n \cap G \neq \emptyset\right) = 1. \quad (7)$$

D11:

$$\bigcup_{j=1}^{\infty} \{\omega \mid \mathbf{x}^j(\omega) \in G\} \subseteq \bigcup_{j=1}^{\infty} \{\omega \mid \mathcal{X}^j(\omega) \cap G \neq \emptyset\}.$$

D12:

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} \mathcal{X}^n \cap G \neq \emptyset\right) = 1. \quad (8)$$

D13:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k) = \mathbf{h}(\mathbf{x}_k) + \mathbf{g}(\mathbf{x}_k), \quad \mathbf{x}_k \in \mathbb{R}^n. \quad (9)$$

D14:

$$|g_i(\mathbf{x}) - g_i(\boldsymbol{\mu})| \leq L_{g_i} \|\mathbf{x} - \boldsymbol{\mu}\|_2, \quad i = 1, \dots, n. \quad (10)$$

D15:

$$\mathcal{B}(\boldsymbol{\mu}, \mathbf{Q}) := \{ \mathbf{x} \mid (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{Q}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \leq 1 \}. \quad (11)$$

D16:

$$\boldsymbol{\mu}_{k+1} = \mathbf{f}(\boldsymbol{\mu}_k). \quad (12)$$

D17:

$$\begin{aligned} \mathbf{x}_{k+1} - \boldsymbol{\mu}_{k+1} &= \mathbf{h}(\mathbf{x}_k) - \mathbf{h}(\boldsymbol{\mu}_k) + \mathbf{g}(\mathbf{x}_k) - \mathbf{g}(\boldsymbol{\mu}_k) \\ &= \nabla \mathbf{h} \cdot (\mathbf{x}_k - \boldsymbol{\mu}_k) + \mathbf{g}(\mathbf{x}_k) - \mathbf{g}(\boldsymbol{\mu}_k). \end{aligned}$$

D18:

$$|g_i(\mathbf{x}_k) - g_i(\boldsymbol{\mu}_k)| \leq L_{g_i} \|\mathbf{x}_k - \boldsymbol{\mu}_k\|_2 \leq L_{g_i} \lambda_{\max}(\mathbf{Q}_k), \quad (13)$$

$$\text{D19:} \quad \mathbf{g}(\mathbf{x}_k) - \mathbf{g}(\boldsymbol{\mu}_k) \in \mathcal{B}(\mathbf{0}, \mathbf{Q}_{\mathbf{g}_k}), \quad \text{where } \mathbf{Q}_{\mathbf{g}_k} = n \cdot \text{diag}((L_{g_i} \lambda_{\max}(\mathbf{Q}_k)^2), i = 1, \dots, n), \quad (14)$$

$$\text{D20:} \quad \mathbf{x}_{k+1} - \boldsymbol{\mu}_{k+1} \in \mathcal{B}(\mathbf{0}, \mathbf{Q}_{\text{nom},k}) \oplus \mathcal{B}(\mathbf{0}, \mathbf{Q}_{\mathbf{g}_k}) \subset \mathcal{B}(\mathbf{0}, \mathbf{Q}_{k+1}), \quad (15)$$

$$\text{D21:} \quad \mathbf{x}_{k+1} \in \mathcal{B}(\boldsymbol{\mu}_{k+1}, \mathbf{Q}_{k+1}). \quad (16)$$

$$\text{D22:} \quad \text{Vol}(\mathcal{B}(\boldsymbol{\mu}, \mathbf{Q})) = \frac{\pi^{n/2}}{\Gamma(n/2 + 2)} \frac{1}{\sqrt{\det(\mathbf{Q}^{-1})}}, \quad (17)$$

$$\text{D23:} \quad \boldsymbol{\mu}_{k+1} = \mathbf{f}(\boldsymbol{\mu}_k, \mathbf{u}_k, \bar{\boldsymbol{\theta}}, \bar{\mathbf{w}}_k), \quad \bar{\boldsymbol{\theta}} \in \Theta, \quad \bar{\mathbf{w}}_k \in \mathbb{W}, \quad (18)$$

$$\text{D24:} \quad \min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} l(\boldsymbol{\mu}_k, \mathbf{u}_k) + l_f(\boldsymbol{\mu}_N) \quad (19a)$$

$$\text{D25:} \quad \text{subject to} \quad \mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\theta}, \mathbf{w}_k), \quad \mathbf{w}_k \in \mathbb{W}, \quad \boldsymbol{\theta} \in \Theta, \quad k = 0, \dots, N-1, \quad (19b)$$

$$\text{D26:} \quad \bigcap_{k=1}^N (\mathbf{x}_k \in \mathcal{X}_{\text{free}}) \cap \bigcap_{k=0}^{N-1} (\mathbf{u}_k \in \mathcal{U}) \cap (\mathbf{x}_N \in \mathcal{X}_{\text{goal}}) \cap (\mathbf{x}_0 \in \mathcal{X}_0). \quad (19c)$$

$$\text{D27:} \quad \min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} l(\boldsymbol{\mu}_k, \mathbf{u}_k) + l_f(\boldsymbol{\mu}_N) \quad \text{s.t.} \quad \bigcap_{k=1}^N \mathcal{X}_k \subset \mathcal{X}_{\text{free}}, \quad \bigcap_{k=0}^{N-1} \mathbf{u}_k \in \mathcal{U}, \quad \mathcal{X}_N \subset \mathcal{X}_{\text{goal}}, \quad (20)$$

$$\text{D28:} \quad \mathbf{x}_{\min,i} \leq \mathbf{x}_{k,i}(\mathbf{u}) \leq \mathbf{x}_{\max,i}, \quad \forall (\mathbf{x}_0, \boldsymbol{\theta}, \mathbf{w}_{0:k-1}) \in \mathcal{X}_0 \times \Theta \times \mathbb{W}^k, \quad (21)$$

$$\text{D29:} \quad \mathbf{x}_{\min,i} + \delta_{k,i}(\mathbf{u}) \leq \boldsymbol{\mu}_{k,i}(\mathbf{u}) \leq \mathbf{x}_{\max,i} - \delta_{k,i}(\mathbf{u}). \quad (22)$$

Inline

- I1: 1
- I2: k
- I3: $\mathcal{X}_k^M(\omega)$
- I4: $\mathcal{X}_k$
- I5: $M \rightarrow \infty$
- I6: 2
- I7: $\mathbb{N}$
- I8: $\mathbb{R}$
- I9: σ
- I10: $\mathbb{R}^n$
- I11: $\mathcal{B}(\mathbb{R}^n)$
- I12: $\mathcal{F}$
- I13: $\mathcal{G}$
- I14: $\mathcal{K}$
- I15: $\mathcal{S}^N := \mathcal{S} \times \dots \times \mathcal{S}$
- I16: N
- I17: $\mathcal{S}$
- I18: $\|\mathbf{x}\|_{\mathbf{Q}}^2 = \mathbf{x}^T \mathbf{Q} \mathbf{x}$
- I19: $\mathbf{Q}$
- I20: $\circ$
- I21: $\mathcal{X}^M = \{\mathbf{x}^j\}_{j=1}^M$

I22: $\text{Co}(\mathcal{X}^M)$
 I23: $\mathbf{x}_k \in \mathbb{R}^n$
 I24: $\mathbf{u}_k \in \mathcal{U}_k$
 I25: $\boldsymbol{\theta} \in \Theta$
 I26: $\mathbf{w}_k \in \mathbb{W}$
 I27: $\mathbf{x}_0 \in \mathcal{X}_0$
 I28: $\mathcal{U}_k \subset \mathbb{R}^m$
 I29: $\Theta \subset \mathbb{R}^p$
 I30: $\mathbb{W} \subset \mathbb{R}^q$
 I31: $\mathcal{X}_0 \subset \mathbb{R}^n$
 I32: $\mathbf{f}(\cdot)$
 I33: $N \in \mathbb{N}$
 I34: $\mathcal{X}_1, \dots, \mathcal{X}_N$
 I35: $k = 1, \dots, N$
 I36: $\mathbf{u} = \boldsymbol{\kappa}(\mathbf{x})$
 I37: $\mathbf{f}(\mathbf{x}_k, \boldsymbol{\kappa}(\mathbf{x}_k), \boldsymbol{\theta}, \mathbf{w}_k)$
 I38: $(\mathbf{u}_0, \dots, \mathbf{u}_{N-1})$
 I39: $\mathcal{U}_i = \{\mathbf{u}_i\}$
 I40: $\mathbf{f}$
 I41: $\mathcal{X}_0$
 I42: $\mathcal{U}_i$
 I43: Θ
 I44: $\mathbb{W}$
 I45: $\boldsymbol{\theta}$
 I46: $\mathbf{x}_{k+1} = \boldsymbol{\theta} \mathbf{x}_k$
 I47: $\boldsymbol{\theta} \in \{1, 2\}$
 I48: $\mathcal{X}_0 = \{1\}$
 I49: $\mathcal{X}_2 = \{1, 4\}$
 I50: $\tilde{\mathcal{X}}_2 = \{1, 2, 4\}$
 I51: n, m, p, q
 I52: $(\mathbf{x}_0, \mathbf{u}, \boldsymbol{\theta}, \mathbf{w}) \in \mathcal{X}_0 \times \mathcal{U}^{k-1} \times \Theta \times \mathbb{W}^{k-1}$
 I53: $\mathbf{x}_k$
 I54: $\mathcal{X}_0, \mathcal{U}, \Theta$
 I55: M
 I56: $\mathbf{x}_0^j$
 I57: $\mathbf{u}^j$
 I58: $\boldsymbol{\theta}^j$
 I59: $\mathbf{w}^j$
 I60: $\mathbf{x}_k^j$
 I61: $\{\mathbf{x}_k^j\}_{j=1}^M$
 I62: $\{\mathbf{x}_0^j, \mathbf{u}^j, \boldsymbol{\theta}^j, \mathbf{w}^j\}_{j=1}^M$
 I63: $\mathbf{x}^j$
 I64: $(\mathbf{x}_0^j, \mathbf{u}^j, \boldsymbol{\theta}^j, \mathbf{w}^j)$
 I65: $K \subset \mathbb{R}^n$
 I66: $\mathcal{X} : \Omega \rightarrow \mathcal{F}$
 I67: $(\Omega, \mathcal{A}, \mathbb{P})$
 I68: $\{\omega \mid \mathcal{X}(\omega) \cap \mathcal{K} \neq \emptyset\} \in \mathcal{A}$
 I69: $K \subset \mathbb{R}^n$
 I70: $(\mathbf{x}_0^j, \mathbf{u}^j, \boldsymbol{\theta}^j, \mathbf{w}^j) \in \mathcal{X}_0 \times \mathcal{U}^N \times \Theta \times \mathbb{W}^{N-1}$
 I71: $\mathcal{U}$
 I72: $\alpha = \beta \ll 1$
 I73: $\mathcal{X}_k^M$
 I74: $j = 1, \dots, M$
 I75: $\mathbf{x}_{1:N}^j \leftarrow \text{Propagate}(\mathbf{x}_0^j, \mathbf{u}^j, \boldsymbol{\theta}^j, \mathbf{w}^j)$
 I76: $\mathcal{X}_k^M = \text{Co}(\{\mathbf{x}_k^j\}_{j=1}^M)$
 I77: $k = 1, \dots, N$

I78: $\{\mathcal{X}_k^M\}_{k=1}^N$
 I79: $\mathbf{f} \in \mathcal{C}^1$
 I80: $\{\mathcal{X}^m\}_{m=1}^\infty$
 I81: $\mathcal{X}$
 I82: $K \in \mathcal{K}$
 I83: $G \in \mathcal{G}$
 I84: $\{(\mathbf{x}_0^j, \mathbf{u}^j, \boldsymbol{\theta}^j, \mathbf{w}^j)\}_{j=1}^m$
 I85: $\mathcal{X}_0 \times \mathcal{U}^{k-1} \times \Theta \times \mathbb{W}^{k-1}$
 I86: $\mathcal{X}_k^m = \text{Co}(\{\mathbf{x}_k^j\}_{j=1}^m)$
 I87: $\mathbb{P}(\mathbf{x}_k^j \in G_k) > 0$
 I88: G_k
 I89: $\mathcal{X}_k \cap G_k \neq \emptyset$
 I90: $m \rightarrow \infty$
 I91: $\mathcal{X}_k^m$
 I92: $\text{Co}(\mathcal{X}_k)$
 I93: $\mathcal{X}^m$
 I94: ϵ
 I95: 3
 I96: $\mathbf{z}^j = (\mathbf{x}_0^j, \mathbf{u}^j, \boldsymbol{\theta}^j, \mathbf{w}^j)$
 I97: $\mathbf{z}^{M+1}$
 I98: $\mathbf{x}_k^{M+1} \in \mathcal{X}_k^M$
 I99: $\mathcal{X}_k^{M+1} = \mathcal{X}_k^M$
 I100: $\mathbf{x}_k^{M+1} \notin \mathcal{X}_k^M$
 I101: $\mathbf{z}^j$
 I102: $\mathbf{z}$
 I103: $\mathbf{c}_k^M$
 I104: $\mathbf{Q}_k^M$
 I105: $\mathbf{x}_k(\mathbf{z}) = \mathbf{f}(\cdot, \mathbf{u}_{k-1}, \boldsymbol{\theta}, \mathbf{w}_{k-1}) \circ \dots \circ \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0, \boldsymbol{\theta}, \mathbf{w}_0)$
 I106: $\mathcal{Z} = \mathcal{X}_0 \times \mathcal{U}^k \times \Theta \times \mathbb{W}^{k-1}$
 I107: $\{\mathbf{z}^j\}_{j=1}^M = (\mathbf{x}_0^j, \mathbf{u}^j, \boldsymbol{\theta}^j, \mathbf{w}^j)_{j=1}^M$
 I108: $\boldsymbol{\eta}$
 I109: n_{adv}
 I110: $\mathcal{X}_N^{n_{\text{adv}}} = \{\mathbf{x}^j\}_{j=1}^{M \cdot n_{\text{adv}}}$
 I111: $\mathbf{x}_{1:N}^j \leftarrow \text{Propagate}(\mathbf{z}^j)$
 I112: $\mathcal{X}_N^0 = \{\mathbf{x}_{1:N}^j\}_{j=1}^M$
 I113: $i = 1, \dots, n_{\text{adv}}$
 I114: $\mathbf{z}^j \leftarrow \mathbf{z}^j + \boldsymbol{\eta} \nabla_{\mathbf{z}} \mathcal{L}^M(\mathbf{x}_{1:N}^j)$
 I115: $\mathbf{z}^j \leftarrow \text{Proj}_{\tilde{\mathcal{Z}}}(\mathbf{z}^j)$
 I116: $\mathcal{X}_N^i \leftarrow \mathcal{X}_N^{i-1} \cup \{\mathbf{x}_{1:N}^j\}_{j=1}^M$
 I117: $\text{Co}(\mathcal{X}_N^{n_{\text{adv}}})$
 I118: $\mathbf{x}_0$
 I119: $\mathbf{u}$
 I120: $\mathbf{w}$
 I121: $\mathcal{L}(\cdot)$
 I122: $(1 - \epsilon)$
 I123: $\mathbf{x}_1 = \mathbf{x}_0 + \mathbf{u}_0, \mathbf{x}_1 \in \mathbb{R}^n, \mathbf{x}_0 \in [-1, 1]^n$
 I124: $\mathbf{u}_0 \in [-\bar{u}, \bar{u}]^n$
 I125: $\eta = 1$
 I126: $n_{\text{adv}} = 1$
 I127: $\mathcal{X}_0 \times \mathcal{U}_0$
 I128: δ
 I129: $\bar{u}$
 I130: $\bar{u} \in \{0.5, 1, 2\}$
 I131: $n \in \{3, 4\}$
 I132: 3σ

I133: $\mathbf{x}_k = (\mathbf{p}_k, \mathbf{v}_k) \in \mathbb{R}^4$
 I134: $\mathbf{u}_k \in \mathbb{R}^2$
 I135: $\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k$
 I136: $\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{u}_k$
 I137: (128, 128)
 I138: $\tanh(\cdot)$
 I139: $\mathcal{L}_2$
 I140: $\mathcal{X}_0 \times \mathcal{U}_k^N$
 I141: $N = 10$
 I142: $\mathcal{U}_k = \{\mathbf{u}_k\}$
 I143: 10^{-7}
 I144: ± 2
 I145: $\mathcal{X}_1$
 I146: $\mathcal{X}_2$
 I147: $\mathcal{X}_4$
 I148: $\mathcal{X}_5$
 I149: $M=3k$
 I150: $\frac{M=2k}{n_{\text{adv}}=1}$
 I151: $\frac{4}{4}$
 I152: $\mathbf{x} = [\mathbf{p}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega}] \in \mathbb{R}^{13}$
 I153: $\mathbf{u} = [\mathbf{F}, \mathbf{M}] \in \mathbb{R}^6$
 I154: $\dot{\mathbf{p}} = \mathbf{v}$
 I155: $m\dot{\mathbf{v}} = \mathbf{F}$
 I156: $\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Omega}(\boldsymbol{\omega})\mathbf{q}$
 I157: $\mathbf{J}\dot{\boldsymbol{\omega}} = \mathbf{M} - \mathbf{S}(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega}$
 I158: $J = \text{diag}([J_x, J_y, J_z])$
 I159: $\Delta t = 5\text{s}$
 I160: $\mathbf{w}_k$
 I161: $\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\theta}_k) + \mathbf{w}_k$
 I162: $m \in [7.1, 7.3]$
 I163: $J_i \in [0.065, 0.075]$
 I164: $|w_{ki}| \leq 10^{-4}$
 I165: $i = 1, \dots, 13$
 I166: $|w_{ki}| \leq 5 \times 10^{-4}$
 I167: $i = 4, 5, 6$
 I168: $M = 100$
 I169: $N = 21$
 I170: 875ms
 I171: $(\mathbf{x}^j, \mathcal{X}^m, \mathcal{X}) = (\mathbf{x}_k^j, \mathcal{X}_k^m, \text{Co}(\mathcal{X}_k))$
 I172: $\mathbf{z} = (\mathbf{x}_0, \mathbf{u}, \boldsymbol{\theta}, \mathbf{w}) \in \mathbb{R}^z$
 I173: $\mathbf{f}(\mathbf{z}) = \mathbf{f}(\cdot, \mathbf{u}_{k-1}, \boldsymbol{\theta}, \mathbf{w}_{k-1}) \circ \dots \circ \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0, \boldsymbol{\theta}, \mathbf{w}_0)$
 I174: $(\Omega, \mathcal{A}, \mathbb{P})$
 I175: $\mathcal{X}^m : \Omega \rightarrow \mathcal{F}$
 I176: $\mathcal{X}^m(\omega) = \text{Co}(\{\mathbf{x}^j\}_{j=1}^m)$
 I177: $\{\mathcal{X}^m(\omega), m \geq 1\}$
 I178: $\mathcal{X}^1(\omega) \subseteq \mathcal{X}^2(\omega) \subseteq \dots$
 I179: $j = 1, \dots, m$
 I180: $\{\mathbf{x}^j\}$
 I181: $\mathcal{X}^m = \text{Co}(\{\mathbf{x}^j\}_{j=1}^m)$
 I182: $\mathcal{X}^m(\omega)$
 I183: $\mathcal{Z}$
 I184: $\mathcal{X}^m \subseteq \mathcal{X}^{m+1}$
 I185: m
 I186: $K \in \mathcal{K}$
 I187: $\mathcal{X} \cap K = \emptyset$
 I188: $\mathbf{x}^j(\omega) \in \mathcal{X}$

I189: j
 I190: $\mathcal{X}^m \subset \text{Co}(\mathcal{X})$
 I191: $\mathcal{X}^m \cap K = \emptyset$
 I192: $\mathbb{P}(\mathcal{X}^m \cap K \neq \emptyset) = 0$
 I193: $\sum_{m=1}^{\infty} \mathbb{P}(\mathcal{X}^m \cap K \neq \emptyset) = 0$
 I194: $\mathbb{P}(\mathcal{X}^m \cap K \neq \emptyset \text{ i.o.}) = 0$
 I195: $G \in \mathcal{G}$
 I196: $\mathcal{X} \cap G \neq \emptyset$
 I197: $\mathbb{P}(\mathcal{X}^m \cap G = \emptyset \text{ i.o.}) = 0$
 I198: G
 I199: $\mathcal{X}^m \subseteq \mathcal{X}^{m+1}$
 I200: $\mathbb{P}(\mathbf{x}^j \in G \text{ i.o.}) = 1$
 I201: $\{\omega \mid \mathbf{x}^j \in G\} = \{\omega \mid \mathbf{z}^j \text{ s.t. } \mathbf{f}(\mathbf{z}^j) \in G\}$
 I202: $\{\omega \mid \mathbf{x}^j \in G\}$
 I203: $j = 1, \dots, m$
 I204: $\mathcal{Z}_G \subset \mathbb{R}^z$
 I205: $\mathbf{z} \in \mathcal{Z}_G$
 I206: $\mathbf{f}(\mathbf{z}) \in G$
 I207: $\mathcal{Z}_G$
 I208: $\mathbb{P}(\mathbf{x}^j \in G) = \mathbb{P}(\mathbf{z}^j \in \mathcal{Z}_G) = \alpha_G > 0$
 I209: α_G
 I210: $(\mathbf{x}_0, \mathbf{u}, \boldsymbol{\theta}, \mathbf{w})$
 I211: $\sum_{j=1}^{\infty} \mathbb{P}(\mathbf{x}^j \in G) = \infty$
 I212: $\mathbb{P}(\mathbf{x}^j \in G \text{ i.o.}) = \mathbb{P}(\cup_{n=1}^{\infty} \cap_{m=n}^{\infty} \mathbf{x}^j \in G) = 1$
 I213: $\cap_{n=1}^{\infty} A_n \subseteq A_1$
 I214: $\mathcal{X}^m \subseteq \mathcal{X}^{m+1}, \forall m$
 I215: $\{\omega \mid \cap_{m=n}^{\infty} \mathcal{X}^m \cap G \neq \emptyset\} = \{\omega \mid \mathcal{X}^n \cap G \neq \emptyset\}$
 I216: $\mathbf{x}^j \in \mathcal{X}^j$
 I217: $\{\omega \mid \mathbf{x}^j(\omega) \in G\} \subseteq \{\omega \mid \mathcal{X}^j(\omega) \cap G \neq \emptyset\}$
 I218: $\{\mathcal{X}^m(\omega), m \geq 1\}$
 I219: $\square$
 I220: $n_{\text{adv}} \geq 1$
 I221: $n_{\text{adv}} \in \{1, \dots, 5\}$
 I222: $M = 100$
 I223: $n_{\text{adv}} = 1$
 I224: $\mathbf{x}_0$
 I225: (M, n_{adv})
 I226: $(100, 1)$
 I227: x, y
 I228: $N = 20$
 I229: $M \in \{50, 100, 200\}$
 I230: $\{83, 173, 304\}$
 I231: $M = 200$
 I232: $n_{\text{adv}} \in \{1, 2, 5\}$
 I233: $k = 12$
 I234: $k = 8, 14$
 I235: $\mathbf{w}_k^j$
 I236: $(m^j, J_i^j) = (7.1, 0.075)$
 I237: $k = 8$
 I238: $(7.3, 0.065)$
 I239: $k = 14$
 I240: $\mathbf{w}_k^j$
 I241: $\mathbf{u}_k$
 I242: $\mathbf{h}$
 I243: $\mathbf{g}$
 I244: $\mathbf{x}, \boldsymbol{\mu} \in \mathbb{R}^n$

I245: $\mathcal{B}(\boldsymbol{\mu}, \mathbf{Q})$
 I246: $\boldsymbol{\mu} \in \mathbb{R}^n, \mathbf{Q} \in \mathbb{R}^{n \times n}, \mathbf{Q} \succ 0$
 I247: $\mathbf{x}_k \in \mathcal{B}(\boldsymbol{\mu}_k, \mathbf{Q}_k)$
 I248: $\boldsymbol{\mu}_{k+1}, \mathbf{Q}_{k+1}$
 I249: $\mathbf{x}_{k+1} \in \mathcal{B}(\boldsymbol{\mu}_{k+1}, \mathbf{Q}_{k+1})$
 I250: $\mathbf{f}(\mathbf{x}_k)$
 I251: $\mathbf{x}$
 I252: $\nabla \mathbf{h} \cdot (\mathbf{x}_k - \boldsymbol{\mu}_k) \in \mathcal{B}(\mathbf{0}, \mathbf{Q}_{\text{nom},k})$
 I253: $\mathbf{Q}_{\text{nom},k} = \mathbf{h} \mathbf{Q}_k \mathbf{h}^T$
 I254: $\lambda_{\max}(\mathbf{Q}_k)$
 I255: $\mathbf{Q}_k$
 I256: $\mathbf{g}(\mathbf{x}_k) - \mathbf{g}(\boldsymbol{\mu}_k)$
 I257: $\text{diag}(\dots)$
 I258: $(\dots)$
 I259: $\mathbf{Q}_{k+1} = \frac{c+1}{c} \mathbf{Q}_{\text{nom},k} + (1+c) \mathbf{Q}_{\mathbf{g}_k}$
 I260: $c = \sqrt{\text{Tr}(\mathbf{Q}_{\text{nom},k}) / \text{Tr}(\mathbf{Q}_{\mathbf{g}_k})}$
 I261: $\text{Tr}(\cdot)$
 I262: $\mathbf{x}_0 \in \mathcal{B}(\boldsymbol{\mu}_0, \mathbf{Q}_0)$
 I263: $k = 0, \dots, N-1$
 I264: $\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) = \mathbf{h}(\mathbf{x}_k) + \mathbf{g}(\mathbf{x}_k, \mathbf{u}_k)$
 I265: $\mathbf{h}(\mathbf{x}_k) = \mathbf{x}_k$
 I266: 2
 I267: 128
 I268: $\{(\mathbf{x}_k, \mathbf{u}_k, \mathbf{x}_{k+1})^b\}_{b=1}^B$
 I269: $\sum_{b=1}^B \|\mathbf{x}_{k+1}^b - \mathbf{g}(\mathbf{x}_k^b, \mathbf{u}_k^b)\|_2^2$
 I270: $B = 20$
 I271: L_2
 I272: 10^{-6}
 I273: 0.02
 I274: $(1 - 10^{-6})$
 I275: $B = 100$
 I276: $\mathbf{x}_0^b \in \mathcal{X}_0 = \mathcal{B}(\boldsymbol{\mu}_0, \mathbf{Q}_0)$
 I277: $\mathbf{u}^b = (\mathbf{u}_0^b, \dots, \mathbf{u}_{N-1}^b)$
 I278: $b = 1, \dots, B$
 I279: $\boldsymbol{\mu}_0^b = (\mathbf{p}_0^b, \mathbf{v}_0^b)$
 I280: $\mathbf{p}_{0,i}^b \sim \text{Unif}(-5, 5)$
 I281: $\mathbf{v}_{0,i}^b \sim \text{Unif}(-1, 1)$
 I282: $\mathbf{Q}_0^b = 10^{-3} \text{diag}([10, 10, 2, 2])$
 I283: $\mathbf{u}_k^b = \bar{\mathbf{u}}^b + \delta \mathbf{u}_k^b$
 I284: $\bar{\mathbf{u}}_i^b \sim \text{Unif}(-0.4, 0.4)$
 I285: $\delta \mathbf{u}_{k,i}^b \sim \text{Unif}(-0.02, 0.02)$
 I286: $\bar{\mathbf{u}}_i^b \sim \text{Unif}(-0.1, 0.1)$
 I287: $\delta \mathbf{u}_{k,i}^b \sim \text{Unif}(-0.005, 0.005)$
 I288: 1k
 I289: .
 I290: 3k
 I291: 10k
 I292: $\frac{1}{1}$
 I293: $\mathbf{h}(\mathbf{x}) = \mathbf{x}$
 I294: $\mathbf{g}(\mathbf{x}, \mathbf{u})$
 I295: $L_{g_i} = 1$
 I296: $i = 1, 2$
 I297: $L_{g_i} = 0$
 I298: $i = 3, 4$
 I299: $\mathcal{B}(\boldsymbol{\mu}, \mathbf{Q})$

I300: $\Gamma(\cdot)$
 I301: $(\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1})$
 I302: $(\mathbf{x}_k \in \mathcal{X}_{\text{free}}, \mathbf{u}_k \in \mathcal{U})$
 I303: $\boldsymbol{\theta} \in \Theta$
 I304: $\mathbf{w}_k \in \mathbb{W}$
 I305: $\mathbf{x}_N$
 I306: $\mathcal{X}_{\text{goal}}$
 I307: $l : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}, l_f : \mathcal{X} \rightarrow \mathbb{R}$
 I308: $\mathbf{u} = (\mathbf{u}_0, \dots, \mathbf{u}_{N-1})$
 I309: $\boldsymbol{\mu} = (\boldsymbol{\mu}_0, \dots, \boldsymbol{\mu}_N)$
 I310: $\boldsymbol{\mu}_0 \in \mathcal{X}_0$
 I311: $\boldsymbol{\theta}$
 I312: $\bar{\mathbf{w}}_k$
 I313: $\sum_{k=0}^{N-1} l(\boldsymbol{\mu}_k, \mathbf{u}_k) + l_f(\boldsymbol{\mu}_N)$
 I314: $\{\mathcal{X}_k\}_{k=0}^N$
 I315: $\mathbf{u}_{0:N-1}$
 I316: $\mathcal{X}_0$
 I317: $(i+1)$
 I318: $(\boldsymbol{\mu}^i, \mathbf{u}^i)$
 I319: $(\boldsymbol{\mu}^{i+1}, \mathbf{u}^{i+1})$
 I320: $\|\boldsymbol{\mu}^{i+1} - \boldsymbol{\mu}^i\| + \|\mathbf{u}^{i+1} - \mathbf{u}^i\|$
 I321: 5
 I322: $i \in \{1, \dots, n\}$
 I323: $\mathbf{x}_k \in \mathcal{X}_{\text{free}}$
 I324: $\mathcal{X}_k(\mathbf{u})$
 I325: $\boldsymbol{\mu}$
 I326: $\Delta_k(\mathbf{u})$
 I327: $\Delta_k = \{\mathbf{x}_k \mid |\mathbf{x}_{k,i} - \boldsymbol{\mu}_{k,i}| \leq \delta_{k,i}\}$
 I328: $\delta_{k,i} = \max_j |\mathbf{x}_{k,i}^j - \boldsymbol{\mu}_{k,i}|$
 I329: $\mathbf{a} \cdot \mathbf{x}_{k,1:s} \leq b$
 I330: s
 I331: $\mathbf{a} \in \mathbb{R}^s, b \in \mathbb{R}$
 I332: $\mathcal{B}^s(\boldsymbol{\mu}_k, \mathbf{Q}_k) = \{\mathbf{x} \mid (\mathbf{x} - \boldsymbol{\mu}_{k,1:s})^T \mathbf{Q}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_{k,1:s}) \leq 1\}$
 I333: $\mathbf{Q}_k = s \cdot \text{diag}(\delta_i^2, i = 1, \dots, s)$
 I334: $\mathcal{B}^s$
 I335: $\mathbf{a}^T \boldsymbol{\mu}_{k,1:s} + (\mathbf{a}^T \mathbf{Q}_k \mathbf{a})^{1/2} \leq b$
 I336: K
 I337: $\mathcal{X}_0 \times \mathcal{U}^N \times \Theta \times \mathbb{W}^{N-1}$
 I338: $i = 1, \dots, k-1$
 I339: $\mathbf{u}_0, \mathbf{w}_0$
 I340: $\mathbf{x}_k(\mathbf{z})$
 I341: $\mathcal{X}_0, \mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_4, \mathcal{X}_5$
 I342: $\boldsymbol{\theta}_k$
 I343: c
 I344: $\Gamma(n/2 + 2)$